Primitive Root

1. Question: Show that 2 is a primitive root module 11.

Ans: Given that,

A number g is a primitive root modulo 11.

Since 11 is prime, we need the order of 2 modulo 11 to be $\varphi(11) = 10$

Compute powers of 2 mod 11

$2^1 = 2$

$2^2 = 4$

$2^3 = 8$

$2^4 \equiv 16 \equiv 5$

$2^5 \equiv 2^4 \cdot 2 = 5 \cdot 2 = 10$

$2^6 = 10 \cdot 2 = 20 \equiv 9$

$2^7 = 9 \cdot 2 = 18 \equiv 7$

$2^8 = 7 \cdot 2 = 14 \equiv 3$

$2^9 \equiv 3 \cdot 2 \equiv 6$

$2^{10} \equiv 6 \cdot 2 = 12 \equiv 1$

We reached 1 first at exponent 10, so the order of 2 mod 11 is $10 = \varphi(11)$

Therefore 2 is a primitive root modulo 11

2. How many incongruent primitive roots does 14 have?

Ans: Primitive roots exists from $n = 2 \cdot 4$ p^k or $2p^k$ with odd prime p. since $14 = 2 \cdot 7$ primitive roots exists.

The number of incongruent primitive roots matulo.

Compute:

$$\phi(k_1) = \phi(2)\ \phi(7) = 1 \cdot 6 = 6$$

So number of primitive roots $= 4(6) = 2$ (Ans)

③ ⓐ Show that,

$$Ord_n(a) = Ord_n(a^{-1})$$

Let $ord_n(a) = k$

That means: $a^k \equiv 1 \pmod{n}$

Now, $(a^k)^{-1} \equiv 1^{-1} \pmod{n}$

Simplify:

$$(a^{-1})^k = 1 \pmod{n}$$

That means the order of a^{-1} divides k

Hence the two orders divid each other so they are equal

$$Ord_n(a) = Ord_n(a^{-1})$$

(b) Yes;

If a is a primitive root mod n,

then $Ord_n(a) = \phi(n)$

From part (a),

$$Ord_n(a^{-1}) = Ord_{(n)}(a) = \phi(n)$$

So, a^{-1} also has order $\phi(n)$

Therefore, a^{-1} is also a primitive root modulo n.

(Ans.)

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